

System Specification & Architecture (Frozen Version)

Final Project Title

Predicting Missing Examination Component Scores: A Comparative Study of Machine Learning Models for High-Stakes Educational Assessment

Project Type

Primary Category:

Regression Project

Secondary Enhancement:

Explainable AI Project

Justification

The system predicts **continuous numerical paper component scores**, making it a regression problem. It also incorporates **SHAP, feature importance, and interpretable visual analytics**, aligning with Explainable AI requirements encouraged by the assignment.

1. Core Problem Statement

High-stakes examinations often involve multiple paper components. In operational or research contexts, one or more paper component scores may be unavailable due to absenteeism, data loss, technical interruptions, or deliberate experimental evaluation.

There is therefore a need for a **transparent, reproducible, and explainable machine learning framework** capable of:

1. Experimentally predicting hidden paper component scores for benchmarking purposes; and
2. Predicting genuinely missing paper component scores for practical operational use.

This project develops a **web-based machine learning application** that predicts missing examination paper component scores using available component scores and compares the predictive performance of multiple machine learning regression models across **2-paper, 3-paper, and 4-paper examination structures**.

2. Final Research Objectives

The study will:

1. Build a **privacy-preserving preprocessing pipeline** for anonymizing confidential examination records before analysis.
2. Support **optional export of anonymized ADA-safe datasets before further processing**.

3. Clean and transform uploaded CSV examination datasets into a machine-learning-ready structure.
4. Support both:
 - **single-subject datasets**
 - **multi-subject datasets**
5. Automatically determine paper count using:

subject code last digit

where available.

6. Allow manual paper-count input when:

subject code is unavailable.

7. Support:

Mode A — Experimental Benchmarking

using deliberate hiding of paper components.

8. Support:

Mode B — Real Missing Score Prediction

using genuine candidate-level missingness.

9. Compare multiple ML regression models.

10. Evaluate model performance using:

- MAE
- MSE
- RMSE
- R^2

11. Apply explainability techniques including:

- SHAP
- feature importance
- partial dependence plots

12. Present findings through an:

executive interactive dashboard.

3. Final Model Set

Core Models (Frozen)

Tree-Based Models

1. **Random Forest Regressor**
2. **Gradient Boosting Regressor**
3. **XGBoost Regressor**
4. **CatBoost Regressor**

Kernel-Based Model

5. **Support Vector Regressor (SVR)**

Optional Bonus Model

Simple Deep Learning Regressor

(using TensorFlow/Keras)

This will only be attempted after the core system is stable.

Reason:

Tabular educational score data often performs strongly with tree-based methods, while deep learning introduces greater complexity.

4. Dual Prediction Modes (Frozen)

The system shall support:

Mode A — ADA / Research Mode

(Experimental Benchmark Mode)

Purpose:

model evaluation and comparison.

Workflow:

Raw WAEC Dataset



Sensitive Field Detection



Anonymization



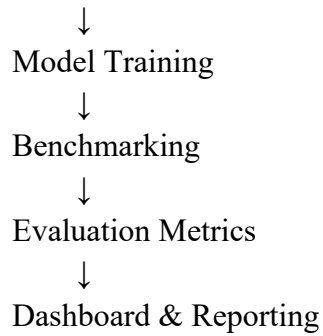
Optional ADA-Safe Dataset Export



Data Cleaning



Experimental Hiding of Paper Components



Important

The anonymized dataset must be exportable:

BEFORE cleaning begins.

This ADA-safe dataset becomes the version submitted to ADA.

Raw WAEC records never leave the protected environment.

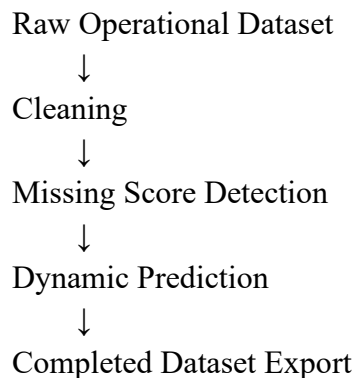
Mode B — Operational Prediction Mode

(Real Missing Score Prediction)

Purpose:

practical prediction of genuinely missing paper components.

Workflow:



Important

In operational mode:

anonymization is optional.

Reason:

Mode B is intended for internal operational use.

5. Privacy-First Data Workflow

Stage 1 — Upload Dataset

User uploads:

raw CSV file

The file may contain:

- Candidate ID
- Centre number
- Subject code
- Subject name
- Paper component scores
- Maximum scores

Stage 2 — Sensitive Field Detection

The application detects:

confidential fields.

Default sensitive fields:

- Candidate ID
- Centre Number
- Candidate Number
- Combined candidate identifiers

Stage 3 — User Confirms Anonymization

User selects anonymization strategy:

Options

- Hash field
- Mask field
- Remove field
- Replace with synthetic ID

Recommended default:

Candidate ID → Synthetic ID

Example:

Raw Candidate ID ADA-safe ID

Raw Candidate ID ADA-safe ID

4380102001 CAND_000001

4380102002 CAND_000002

Stage 4 — Optional ADA-Safe Export

User may click:

Export Anonymized Dataset

This exported file becomes:

ADA submission-safe data.

Stage 5 — Proceed to Cleaning

Only anonymized data enters:

Mode A workflow.

6. Data Cleaning Specification

The app must clean uploaded CSV files into a canonical structure.

Required Canonical Fields

Field	Purpose
anonymized_candidate_id	Anonymous identifier
subject_code	Optional but preferred
subject_name	Optional
paper_count	Required or inferred
p1_score	Paper 1 score
p2_score	Paper 2 score
p3_score	Paper 3 score
p4_score	Paper 4 score
p1_max	Maximum P1
p2_max	Maximum P2
p3_max	Maximum P3
p4_max	Maximum P4

Subject Rules

If subject code exists:

Infer:

paper count from last digit.

Example:

302002 → 2 papers

718003 → 3 papers

402004 → 4 papers

If subject code is missing:

Use:

subject name

and request:

paper count via UI.

If maximum scores are missing:

UI requests:

max score for each paper component.

When multiple subjects exist:

Handle:

subject-by-subject.

7. Scenario Design

Mode A — Experimental Benchmarking

2-paper subjects

- Hide P1 → predict P1
- Hide P2 → predict P2

3-paper subjects

- Hide P1 → predict P1
- Hide P2 → predict P2
- Hide P3 → predict P3

4-paper subjects

- Hide P1 → predict P1
- Hide P2 → predict P2
- Hide P3 → predict P3
- Hide P4 → predict P4

Mode B — Real Missing Prediction (Lite)

Capabilities:

- ✓ Detect genuine missing paper components
- ✓ Candidate-level missingness

Example:

Candidate	Missing Paper
A	P1
B	P3
C	P2

- ✓ Dynamically select correct model
- ✓ Predict missing paper
- ✓ Export completed dataset
- ✓ Add:

prediction_status

Example:

candidate_id	p1	p2	p3	prediction_status
C001	61	53	71	predicted

Deferred to WAEC Version 2

Postpone:

- ✗ Multiple Excel tables per centre
- ✗ Subject-wise workbook tabs
- ✗ Centre-subject routing logic

✗ Operational formatting automation

This protects the 2-week timeline.

8. Feature Engineering

The system should generate:

- normalized scores
- partial total
- mean available score
- score range
- standard deviation
- paper score ratios

This supports:

novel feature engineering bonus marks.

9. Evaluation Framework

Metrics:

- MAE
- MSE
- RMSE
- R^2

Output:

Subject	Scenario	Model	MAE	MSE	RMSE	R^2	Rank
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Best model selected:

per scenario

and

overall.

10. Explainability

Use:

- SHAP
- Feature Importance

- Partial Dependence Plots
- Residual plots
- Actual vs Predicted plots

Supports:

Interpretation & Discussion section.

11. System Architecture

Recommended Stack

Frontend

React + TypeScript

Backend

FastAPI

ML Engine

Python + scikit-learn + XGBoost + CatBoost + SHAP

Visualization

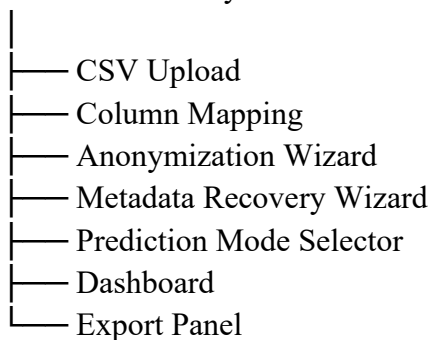
Plotly

Deployment

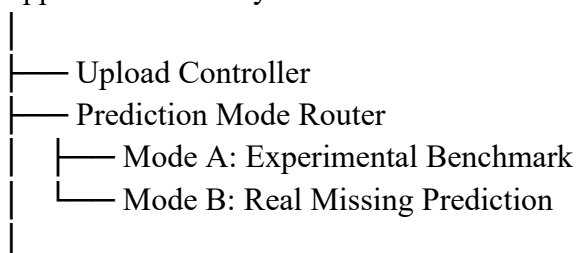
Local demo / Render / Railway / Hugging Face Spaces

Architecture Layers

User Interface Layer



Application/API Layer



- Anonymization Service
- Data Cleaning Service
- Subject Grouping Service
- Scenario Generator
- Model Training Controller
- Evaluation Controller
- Report Export Controller

Machine Learning Layer

- Feature Engineering
- Model Registry
- Training Pipeline
- Cross Validation
- Prediction Engine
- Metrics Engine
- Explainability Engine

Data Layer

- Raw Uploaded File
- ADA-Safe Anonymized Dataset
- Cleaned Dataset
- Prediction Results
- Model Metrics
- Exported Reports

12. Updated Coding-Agent Requirements

The coding agent must build:

- (1) CSV upload system.
- (2) Column mapping UI.
- (3) Sensitive-field anonymization before cleaning.
- (4) Optional ADA-safe export before processing.
- (5) Dual prediction modes.
- (6) Subject detection and grouping.
- (7) Subject-code paper-count inference.
- (8) Manual paper-count recovery.
- (9) Missing maximum-score wizard.
- (10) Single-subject and multi-subject handling.

- (11) 2-, 3-, and 4-paper processing.
- (12) Canonical cleaned dataset generation.
- (13) Feature engineering.
- (14) Scenario generator.
- (15) ML model training.
- (16) Model comparison.
- (17) SHAP explainability.
- (18) Interactive executive dashboard.
- (19) User-selectable charts.
- (20) Mode B Lite missing-score prediction.
- (21) Completed prediction export.
- (22) requirements.txt
- (23) README.md
- (24) Modular, commented, reproducible code.

13. Report Structure Preview

- (1) Introduction
- (2) Short Literature Review
- (3) Data Collection and Description
- (4) Data Privacy and Anonymization
- (5) Data Cleaning and Preprocessing
- (6) Exploratory Data Analysis
- (7) Machine Learning Modelling
- (8) Model Evaluation and Comparison
- (9) Explainability and Interpretation
- (10) Discussion
- (11) Conclusion and Recommendations
- (12) Reproducibility Guide
- (13) References

Dataset Strategy + Experimental Design (Frozen Version)

1. Dataset Strategy

We will support:

A. Single Subject Dataset

Example:

Biology only

or

Mathematics only

Purpose:

Cleaner experiments.

Faster benchmarking.

Better debugging.

Recommended for first testing.

B. Multi-Subject Dataset

Example:

Subject

Biology

Chemistry

Mathematics

System automatically:

groups by subject

then runs independent ML pipelines.

This improves scalability.

Supported Subject Types

2-paper subjects

Example:

P1 + P2

3-paper subjects

Example:

$P1 + P2 + P3$

4-paper subjects

Example:

$P1 + P2 + P3 + P4$

Paper count determined by:

Priority Order

Rule 1

Subject code last digit.

Example:

718003 → 3 papers

402004 → 4 papers

Rule 2

Manual UI input if:

subject code missing.

2. Accepted CSV Structures

The system must accept multiple formats.

Format A — Fully Structured

Example:

subject_code subject_name candidate_id p1 p2 p3

Best-case format.

Format B — Missing Subject Code

Example:

subject_name candidate_id p1 p2 p3

System asks:

paper count.

Format C — Missing Maximum Scores

Example:

p1 p2 p3

System requests:

maxima via UI.

Format D — Multi-Subject File

Example:

subject_code candidate_id p1 p2 p3

Grouped automatically.

3. Privacy Strategy

Mode A (ADA Research)

Mandatory anonymization.

Workflow:

Upload



Sensitive Field Detection



Anonymization



Optional ADA Export



Cleaning

Sensitive fields:

- candidate_id
- centre_no
- any direct identifiers

Replacement:

candidate_000001

No reversibility.

Mode B (Operational)

No anonymization required.

Reason:

Internal use.

4. Canonical Internal Dataset Schema

After cleaning, ALL data becomes:

Field	Type
anonymized_candidate_id	string
subject_code	string
subject_name	string
paper_count	int
p1_score	numeric
p2_score	numeric
p3_score	numeric
p4_score	numeric
p1_max	numeric
p2_max	numeric
p3_max	numeric
p4_max	numeric

This becomes:

ML-ready dataset.

5. Missing Value Rules

The app must recognize:

AB

ABS

A

NaN

-99

blank

missing

Mode A Rule

Experimental hiding only.

No real missingness allowed during benchmarking.

Reason:

Need ground truth.

Mode B Rule

Exactly:

one missing paper allowed per candidate.

Example:

VALID:

P1 P2 P3

61 54 missing

INVALID:

P1 P2 P3

missing missing 70

Reason:

Insufficient predictors.

System warning:

Insufficient available papers for reliable prediction.

This keeps research defensible.

6. Data Cleaning Strategy

The app must automatically perform:

Duplicate Detection

Remove duplicate candidate rows.

Type Correction

Convert:

"45"

to numeric.

Score Validation

Check:

score $\leq$ maximum

Example:

$P1 = 58$

$P1_{\max} = 40$

Reject.

Outlier Detection

Use:

IQR method

BUT:

flag only.

Do not automatically remove.

Reason:

High performers are not errors.

Missing Metadata Recovery

If:

max scores missing →

ask user.

If:

paper count missing →

ask user.

7. Feature Engineering Strategy (Frozen)

This is important.

We freeze features.

Feature 1 — Normalized Score

$$\text{Normalized Score} = \frac{\text{Paper Score}}{\text{Maximum Score}}$$

Critical.

Prevents score-scale bias.

Feature 2 — Partial Total

Example:

Predicting P3:

Feature:

$P1 + P2$

Feature 3 — Mean Available Score

Example:

$(P1 + P2)/2$

Feature 4 — Score Spread

Example:

$\max(\text{score}) - \min(\text{score})$

Measures consistency.

Feature 5 — Standard Deviation

Candidate variability.

Feature 6 — Score Ratios

Example:

$P1/P2$

Optional.

8. Experimental Design (Frozen)

This is the research core.

Mode A — Experimental Benchmarking

The system deliberately hides scores.

Scenario A1

Hide:

P1

Predict:

P1

Scenario A2

Hide:

P2

Predict:

P2

Scenario A3

Hide:

P3

Predict:

P3

Scenario A4

Hide:

P4

Predict:

P4

Only if applicable.

Multiple Subject Logic

Example:

Biology:

Run scenarios.

Chemistry:

Run scenarios.

Mathematics:

Run scenarios.

Then compare.

9. Model Training Strategy

Frozen models:

- (1) Random Forest Regressor
- (2) Gradient Boosting Regressor
- (3) XGBoost Regressor
- (4) CatBoost Regressor
- (5) Support Vector Regressor

Optional:

(6) Deep Learning Regressor

10. Validation Strategy (Frozen)

We will NOT use simple train-test split only.

Use:

80/20 split

PLUS

5-Fold Cross Validation

Why?

Better reproducibility.

More publication-worthy.

11. Evaluation Framework

Metrics:

MAE

MSE

RMSE

R²

Model Ranking Table

Model RMSE MAE R² Rank

Statistical Comparison

Optional but strong:

paired comparison of prediction errors.

Publication bonus.

12. Explainability Strategy

We freeze:

SHAP

Mandatory.

Feature Importance

Mandatory.

Partial Dependence Plots

Mandatory.

Actual vs Predicted

Mandatory.

Residual Analysis

Mandatory.

13. Mode B Lite Experimental Design

Real missing-score prediction.

Workflow:

Upload dataset →

Detect missing paper →

Choose correct trained model →

Predict →

Mark:

prediction_status

Export.

Example:

candidate	p1	p2	p3	status
C001	61	53	71	predicted

14. Export Strategy

Mode A

Export:

- anonymized ADA dataset
- cleaned dataset
- model results
- metrics
- plots

Mode B Lite

Export:

- completed prediction file
- prediction status
- model used

15. Deliverables from Phase 2

We have now frozen:

- ✓ Dataset strategy
- ✓ CSV acceptance rules
- ✓ Privacy workflow
- ✓ Canonical schema
- ✓ Missing value rules
- ✓ Feature engineering
- ✓ Experimental scenarios
- ✓ Validation strategy
- ✓ Evaluation metrics
- ✓ Explainability strategy
- ✓ Export logic
- ✓ Mode B Lite logic